MULTIVARIABLE CALCULUS

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Contents

0.1 intro

summary notes of Larson's *Calculus (10th edition)* on specifically the multivariable chapters, 12-15. these are NOT meant as educational notes, more just a brief overview to remember what the things actually are.

1 Vector-Valued Functions

1.1 Vector-Valued Functions

Vector-valued functions (VVFs) take in a scalar factor and return a vector from it and can always be expressed as $\vec{r}(t) = f(t)\hat{\imath} + g(t)\hat{\jmath}$. Limits and continuity are defined based on the limits of the constituent functions f and g.

1.2 Differentiation and Intergration of Vector-Valued Functions

Differentiation and intergration of VVFs are done component-wise. Constant factor from intergration can be expressed as single vector $\vec{C}$.

Most of the properties of differentiation of VVFs are pretty obvious, except for the "product rule" also applying to the cross product and the dot product. And if $\vec{r}(t) \cdot \vec{r}(t) = c$, then $(\vec{r}(t))' \cdot \vec{r}(t) = 0$.

1.3 Velocity and Acceleration

For a given VVF $\vec{r}(t)$, velocity $= (\vec{r}(t))'$ and acceleration $= (\vec{r}(t))''$.

1.4 Tangent Vectors and Normal Vectors

If C is a smooth curve (continuous velocity + non-zero on the interval), then the **unit tangent vector** $\vec{T}(t)$ is defined as

$$\vec{T}(t) = \frac{(\vec{r}(t))'}{|(\vec{r}(t))'|}$$

Note the unit tangent vector will flip directions depending on the orientation of the curve.

From the constant dot product derivative property mentioned in "Differentation and Intergration of VVFs," the derivative of the unit tangent vector is perpendicular to the curve and results in the **principal unit normal vector**:

$$\vec{N}(t) = \frac{(\vec{T}(t))'}{|(\vec{T}(t))'|}$$

It can be proven that the acceleration vector will always lie in the plane created by $\vec{T}$ and $\vec{N}$, and thus can be expressed as a linear combination of the two:

$$\vec{a}(t) = a_T\vec{T}(t) + a_N\vec{N}(t)$$

where

$$a_T = \vec{a} \cdot \vec{T} = \frac{\vec{v} \cdot \vec{a}}{|\vec{v}|}$$

$$a_N = \vec{a} \cdot \vec{N} = \frac{|\vec{v} \times \vec{a}|}{|\vec{v}|} = \sqrt{|\vec{a}|^2 - a_T^2}$$

Note the dot product definition of the coeffecients comes from the geometry of lying on the plane.

1.5 Arc Length and Curvature

The **arc length** of a curve given by a VVF on an interval $[a, b]$ is defined as the following:

$$s = \int_a^b |(\vec{r}(t))'| \, \mathrm{d}t$$

The **arc length parameter** $s(t)$ is defined as the arc length along the interval $[a, t]$ for variable t. VVFs may be re-written to use the arc length parameter instead of a primary variable, which leads to the very nice property that $|(\vec{r}(s))'| = 1$ for all s. The converse is also true: if t is any parameter in which $|(\vec{r}(t))'| = 1$ for all t, then t is the arc length parameter.

Curvature K, approximately "how much does the curve bend at this point," can be calculated using the arc length parameter:

$$K = \left| \frac{\mathrm{d}\vec{T}}{\mathrm{d}s} \right| = |(\vec{T}(s))'|$$

You can use curvature to create a general equation for the acceleration:

$$\vec{a}(t) = \frac{\mathrm{d}^2 s}{\mathrm{d}t^2} \vec{T} + K \left(\frac{\mathrm{d}s}{\mathrm{d}t} \right) \vec{N}$$

2 Functions of Several Variables

2.1 Introduction to Functions of Several Variables

they sure exist

2.2 Limits and Continuity

In 1d calc, we defined limits based on the distance between two x-values. Similarly, in multivariable calc, we can define limits based on the euclidean distance betwen two points. For 2d, we consider the δ **neighborhood** around a point (x_0, y_0) to be a disc with radius δ:

$$\{(x, y) : \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta\}$$

Hence, we can define the limit of a 2d function as "for every $\epsilon > 0$, we can find a circle around the point with radius δ such that for all points (x, y) within the circle $|f(x, y) - L| < \epsilon$." Note this definition implicitly requires all possible "paths" to approach the same value.

Continuity can thus be defined as

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y) = f(x_0, y_0)$$

for all points within a region R.

Similar approaches can be used to define limits and continuity for N-dimensional functions.

2.3 Partial Derivatives

Partial Derivatives of a multivariable function represent how the function changes with respect to ONE of its independent variables. For example, first partial derivative of f with respect to x:

$$\frac{\partial f}{\partial x} = f_x(x,y) = \lim_{\Delta x \to 0} \frac{f(x+\Delta x, y) - f(x,y)}{\Delta x}$$

For calculation, this effectively means that other variables than the one being derived can be treated as effective constants.

Higher order partial derivatives exist by deriving multiple times. While the notation for deriving to the same variable again is as expected (f_{xx}, $\frac{\partial^2 f}{\partial x^2}$), deriving with respect to another variable is denoted

$$f_{xy} = \frac{\partial^2 f}{\partial x\,\partial y}$$

A partial derivative that involves deriving two different variables is known as a **mixed partial derivative**.

If all the mixed partial derivatives are continuous, then the order of differentiation is irrelevant:

$$f_{xy} = f_{yx}$$

2.4 Differentials

Recall for single variables, we defined the differential as

$$\mathrm{d}f = f'(x)\,\mathrm{d}x$$

We can extend this definition to develop **the total differential**. For a function $f(x_1, \ldots x_n)$,

$$\mathrm{d}f = \sum_{i=1}^{n} \frac{\partial f}{\partial x_i}\,\mathrm{d}x_i$$

We call a multivariable function **differentiable** at a point (x_0, y_0) if $\Delta z = f(x+\Delta x, y+\Delta y) - f(x,y)$ can be rewritten as

$$\Delta z = \frac{\partial f}{\partial x}\Delta x + \frac{\partial f}{\partial y}\Delta y + \epsilon_1 \Delta x + \epsilon_2 \Delta y$$

evaluated at the point, where $(\epsilon_1, \epsilon_2) \to (0,0)$ as $(\Delta x, \Delta y) \to (0,0)$.

Continuity of $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ in a region implies differentiability in that region. Differentiability also implies continuity.

2.5 Chain Rule for Multivariable Functions

If you have $w = f(x_1, x_2, \ldots x_n)$ where each x_i is a differentiable function of $t_1, t_2, \ldots t_m$, then

$$\frac{\partial w}{\partial t_i} = \sum_{j=1}^{n} \frac{\partial w}{\partial x_j} \frac{\partial x_j}{\partial t_i}$$

Logically, how w changes with respect to a non-input variable is the sum of how w changes with respect to one of its input variables multiplied by how that input variable changes with respect to the non-input variable given.

2.6 Directional Derivatives and Gradients

The **directional derivative** is defined for a unit direction vector $\vec{u} = \hat{\imath}\cos\theta + \hat{\jmath}\sin\theta$:

$$D_{\vec{u}} f(x,y) = \lim_{t\to 0} \frac{f(x + t\cos\theta, y + t\sin\theta) - f(x,y)}{t} = \frac{\partial f}{\partial x}\cos\theta + \frac{\partial f}{\partial y}\sin\theta$$

The directional derivative measures the rate of change of the function in a specific direction. Note in the cases $\theta = 0$ and $\theta = \frac{\pi}{2}$ the directional derivative degenerates to $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ respectively.

The **gradient** of f is defined as follows:

$$\nabla f(x,y) = \frac{\partial f}{\partial x}\hat{\imath} + \frac{\partial f}{\partial y}\hat{\jmath}$$

Note this implies

$$D_{\vec{u}} = \nabla f(x,y) \cdot \vec{u}$$

The direction of the gradient gives the direction of greatest increase (other directions would cancel and result in less than maximal), and opposite direction of the gradient gives the direction of least increase.

The definition of the gradient extends to more than 2 variables, and the directional derivative is defined relative to the second definition.

2.7 Tangent Planes and Normal Lines

Planes tangent to a point P can be constructed by constructing a plane that passes through P and is normal to the gradient at P. The line through P in the direction of the gradient is known as the normal line at that point.

2.8 Extrema of Functions of Two Variables

In two dimensions, a point (x_0, y_0) is considered a relative maximum of a function if $f(x_0, y_0) \geq f(x,y)$ for all (x,y) in an open disk around (x_0, y_0). Relative minima are defined similarly.

Critical points are points where the gradient is zero OR at least one partial derivative is zero. Like 1d, relative extrema only occur at critical points.

Similar to the second derivative test, the **second partials test** can be used to determine the status of a critical point:

Definition 2.1 (Second Partials Test)
Let f have continuous second partial derivatives in an open region containing the point (x_0, y_0) s.t. the gradient is zero at the point. Consider

$$d = f_{xx}f_{yy} - (f_{xy})^2$$

where all partial derviatives are evaluated at the critical point.

1. If $d > 0$ and $f_{xx} > 0$, then the critical point is a relative minimum.
2. If $d > 0$ and $f_{xx} < 0$, then the crticial point is a relative maximum.
3. If $d < 0$, then the critical point is a **saddle point** (neither maxiumum nor minimum)
4. If $d = 0$, the test is inconclusive.

The proof of this has to do something with the multidimensional analog of a taylor polynomial.

2.9 Applications of Extrema

One interesting application of 2d extrema is its use in deriving regression functions. By minimizing the squared error $S = \sum_{i=1}^{n}(f(x_i) - y_i)^2$, explicit formulas can be derived for regression variables.

2.10 Lagrange Multipliers

Suppose you want to maximize the function $f(x, y)$ given its inputs must satisfy the constraint $g(x, y) = c$. The maximum and minimum values of f subject to the constraint correspond to contour lines of f that are TANGENT to the contour representing $g(x, y) = c$. Because the gradient of a point is always perpendicular to any contour line, this can be expressed as

$$\nabla f = \lambda_0 \nabla g$$

where λ_0 represents a proportionality constant.

Example 2.2
Find the maximum area of a rectangle inscribed within the ellipse $(x^2/3^2)+(y^2/4^2) = 1$.

Answer 1

The area of any rectangle with a point within the first quadrant can be expressed as $A(x, y) = 4xy$.

$$\nabla A = 4y\hat{i} + 4x\hat{j}$$

Let the constraint be $c(x, y) = (x^2/3^2) + (y^2/4^2) = 1$.

$$\nabla c = \frac{2}{9}x\hat{i} + \frac{1}{8}y\hat{j}$$

From the Lagrange multiplier equation, this leads to the following system of equations:

$$4y = \frac{2}{9}\lambda x$$
$$4x = \frac{1}{8}\lambda y$$
$$\frac{x^2}{9} + \frac{y^2}{16} = 1$$

This can be solved for x, y to yield the maximum value of f.

This can be explicitly codified using the **Lagrangian** $\mathcal{L}$:

$$\mathcal{L}(x, y, \lambda) = f(x, y) + \lambda(g(x, y) - c)$$

More constraints can be added like the following:

$$\mathcal{L}(x, y, z, \lambda, \mu) = f(x, y, z) + \lambda(g(x, y, z) - c_1) + \mu(h(x, y, z) - c_2)$$

Using the Lagrangian, the method of lagrange multipliers is equivalent to

$$\nabla\mathcal{L} = \vec{0}$$